F79SP Session 24: FDEs and BDEs

FDEs and BDEs

We discussed on how to obtain the transition rate matrix Q from $P(t)$. Conversely, how can we obtain $P(t)$ from Q? This can be done via one of two possible systems of differential equations.

Theorem Given a transition rate matrix Q. we have

$$p'_{ij}(t) = \sum_{k \in S} p_{ik}(t) q_{kj} \qquad \text{for all } t \text{ and all } i, j \in S \tag{FDE}$$

—the **(Kolmogorov) forward differential equations**, and also

$$p'_{ij}(t) = \sum_{k \in S} q_{ik} p_{kj}(t) \qquad \text{for all } t \text{ and all } i, j \in S \tag{BDE}$$

—the **(Kolmogorov) backward differential equations**.

Proof. **(S finite.)** Recall the Chapman-Kolmogorov equations:

$$p_{ij}(s+t) = \sum_{k \in S} p_{ik}(s) p_{kj}(t), \qquad i, j \in S, \quad s, t > 0.$$

Differentiate with respect to t to obtain

$$p'_{ij}(s+t) = \sum_{k \in S} p_{ik}(s) p'_{kj}(t),$$

and put $t = 0$ to obtain (FDE) (with s instead of t).

Similarly, differentiate with respect to s to obtain

$$p'_{ij}(s+t) = \sum_{k \in S} p'_{ik}(s) p_{kj}(t),$$

and put $s = 0$ to obtain (BDE). □

Remarks:

- When S is finite, given Q, the transition probabilities $(p_{ij}(\cdot))_{i,j \in S}$ are **uniquely** determined by either FDEs or BDEs, together with the initial condition $p_{ij}(0) = \delta_{ij}$.
- In general the FDEs are more useful in practice for determining the functions $(p_{ij}(\cdot))_{i,j \in S}$. The BDEs are mostly used to prove theorems.

Example: (Modified from T8Q1) Consider a Markov process with state space $S = \{1, 2, 3\}$ and transition rate matrix: $Q = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$. Find $p_{ij}(t)$, for every $i, j \in S$.

- First, note that the symmetry property of Q implies that hence $p_{ii}(t)$ is the same for all $i \in S$, and $p_{ij}(t)$ too for $i \neq j$.
- Referring to FDEs with $i = j = 1$:

$$p'_{11}(t) = \sum_{k \in S} p_{1k}(t) q_{k1}$$
$$p'_{11}(t) = p_{11}(t)q_{11} + p_{12}(t)q_{21} + p_{13}(t)q_{31}$$

TBC

Example: general two-state process. Here we have $S = \{0, 1\}$, where, for example 0 might be **healthy** and 1 might be **sick**—in this very simple model nobody ever dies! The transition rate matrix Q is given by

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

for some $\lambda, \mu > 0$.

The forward differential equations (FDE) for the initial state i become

$$\begin{aligned} p'_{i0}(t) &= p_{i0}(t)(-\lambda) + p_{i1}(t)(\mu) \\ p'_{i1}(t) &= p_{i0}(t)(\lambda) + p_{i1}(t)(-\mu) \end{aligned}$$

and we also have the initial condition $p_{ij}(0) = \delta_{ij}$.

Take $i = 0$: it is easy to verify that

$$\begin{aligned} p_{00}(t) &= \frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t} \\ p_{01}(t) &= \frac{\lambda}{\lambda+\mu} - \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t} \qquad (= 1 - p_{00}(t)) \end{aligned}$$

satisfy the forward differential equations and initial condition, and hence give the transition probabilities.

Take $i = 1$: we similarly obtain (e.g. by symmetry)

$$\begin{aligned} p_{10}(t) &= \frac{\mu}{\lambda+\mu} - \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t} \\ p_{11}(t) &= \frac{\lambda}{\lambda+\mu} + \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t} \qquad (= 1 - p_{10}(t)). \end{aligned}$$

Note also that

$$P(t) \to \begin{pmatrix} \frac{\mu}{\lambda+\mu} & \frac{\lambda}{\lambda+\mu} \\ \frac{\mu}{\lambda+\mu} & \frac{\lambda}{\lambda+\mu} \end{pmatrix} \quad \text{as } t \to \infty,$$

so that the limiting distribution is independent of the initial state.

Recall that the following theorem shows how to obtain the stationary distribution directly from the transition rate matrix Q.

Theorem. A distribution $\boldsymbol{\pi}$ is stationary if and only if

$$\boldsymbol{\pi}Q = \mathbf{0} \qquad (\sum_{i\in S} \pi_i q_{ij} = 0 \quad \text{for all } j).$$

Proof. We prove the result in the case where S is finite.

$$\begin{aligned}
\boldsymbol{\pi} \text{ stationary} &\Rightarrow \sum_{i\in S} \pi_i p_{ij}(t) = \pi_j \qquad \text{for all } j \text{ and for all } t \\
&\Rightarrow \sum_{i\in S} \pi_i p'_{ij}(t) = 0 \qquad \text{for all } j \text{ and for all } t \\
&\Rightarrow \sum_{i\in S} \pi_i q_{ij} = 0 \qquad \text{for all } j \qquad (\text{putting } t = 0)
\end{aligned}$$

Conversely, suppose that $\sum_{i\in S} \pi_i q_{ij} = 0$ for all $j \in S$. Recall the *backward differential equations* (BDE):

$$p'_{ij}(t) = \sum_{k\in S} q_{ik} p_{kj}(t) \qquad \text{for all } i, j \in S.$$

Then, for all $j \in S$,

$$\begin{aligned}
\sum_{i\in S} \pi_i p'_{ij}(t) &= \sum_{i\in S}\sum_{k\in S} \pi_i q_{ik} p_{kj}(t) \\
&= \sum_{k\in S} \left(\sum_{i\in S} \pi_i q_{ik} \right) p_{kj}(t) \\
&= 0 \qquad \text{for all } j.
\end{aligned}$$

Integrate both sides of $\sum_{i\in S} \pi_i p'_{ij}(t) = 0$ w.r.t. t gives:

$$\sum_{i\in S} \pi_i p_{ij}(t) = c.$$

To find c, let $t = 0$, which we know that $p_{jj}(0) = 1$ and $p_{ij}(0) = 0$ for $i \neq j$. This gives $c = \pi_j$ and we yeild the stationarity equation $\sum_{i\in S} \pi_i p_{ij}(t) = \pi_j$. □